

symcoder round-trip demo

electrons=(a,b), muons=(p)

1 Setup

Particle types: electrons = $\{a, b\}$, muons = $\{p\}$

Group G: $G = S_{electrons} \times S_{muons} = S_2 \times S_1$, order = 2

Operations: **mag**($\mathbf{x}$) = $|\mathbf{x}|$ (rank 1, symmetric)

dot($\mathbf{x}, \mathbf{y}$) = $\mathbf{x} \cdot \mathbf{y}$ (rank 2, symmetric)

eps($\mathbf{x}, \mathbf{y}, \mathbf{z}$) = $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (rank 3, antisymmetric)

Event vectors (3-D):

a: $\mathbf{a} = (1.20, 0.30, 0.50)$

b: $\mathbf{b} = (0.40, 1.10, 0.20)$

p: $\mathbf{p} = (0.80, 0.30, 0.90)$

2 Phase 1 Encoding (one orbit per FlavouredOperator in repS)

5 FlavouredOperators in repS

2.1 FO: mag flavour=(1, 0)

Canonical rep: $|\mathbf{a}|$

Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $|\mathbf{a}|, |\mathbf{b}|$

Eval values: +1.3342, +1.1874

Encoded [2 reals]: +1.1874, +1.3342

2.2 FO: mag flavour=(0, 1)

Canonical rep: $|\mathbf{p}|$

Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $|\mathbf{p}|$

Eval values: +1.2410

Encoded [1 reals]: +1.2410

2.3 FO: dot flavour=(2, 0)

Canonical rep: $\mathbf{a} \cdot \mathbf{b}$

Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{b}$

Eval values: +0.9100

Encoded [1 reals]: +0.9100

2.4 FO: dot flavour=(1, 1)

Canonical rep: $\mathbf{a} \cdot \mathbf{p}$

Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{p}, \mathbf{b} \cdot \mathbf{p}$

Eval values: +1.5000, +0.8300

Encoded [2 reals]: +0.8300, +1.5000

2.5 FO: eps flavour=(2, 1)

Canonical rep: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$

Encoder: HalfSortEncoder — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$

Eval values: +0.6760

Encoded [1 reals]: +0.6760

3 Phase 2 Encoding (OverlapBlocks, complementarity-drop disabled)

3.1 Block: dot[(1, 1)] $\times$ dot[(1, 1)]

PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=2 **PairFlavour** overlap= (0, 1), type=11_sigma, kind=ASSOC, encoded dim= 2

Encoded values: +3.2951, +2.9389

Decoded pairs (u, v) :

$(\mathbf{a} \cdot \mathbf{p}, \mathbf{b} \cdot \mathbf{p}) \rightarrow (+0.8300, +1.5000)$

$(\mathbf{b} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{p}) \rightarrow (+1.5000, +0.8300)$

PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (1, 1), type=null_self, kind=NULL_SELF, encoded dim= 0

Decoded pairs (u, v) :

$(\mathbf{a} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{p}) \rightarrow (+0.8300, +0.8300)$

$(\mathbf{b} \cdot \mathbf{p}, \mathbf{b} \cdot \mathbf{p}) \rightarrow (+1.5000, +1.5000)$

3.2 Block: dot[(1, 1)] $\times$ dot[(2, 0)]

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 4

Encoded values: +2.3300, +1.8200, +0.4169, +2.1203

Decoded pairs (u, v) :

$(\mathbf{a} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{b}) \rightarrow (+1.5000, +0.9100)$

$$(\mathbf{b} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{b}) \rightarrow (+0.8300, +0.9100)$$

3.3 Block: $\text{dot}[(1, 1)] \times \text{eps}[(2, 1)]$

PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (1, 1), type=11, kind=ASSOC, encoded dim= 4

Encoded values: +2.3300, +0.0000, +1.7020, -0.4529

Decoded pairs (u, v) :

$$(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (+1.5000, +0.6760)$$

$$(\mathbf{b} \cdot \mathbf{p}, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (+0.8300, -0.6760)$$

3.4 Block: $\text{dot}[(1, 1)] \times \text{mag}[(0, 1)]$

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 4

Encoded values: +2.3300, +2.4819, -0.2950, +2.8915

Decoded pairs (u, v) :

$$(\mathbf{a} \cdot \mathbf{p}, |\mathbf{p}|) \rightarrow (+1.5000, +1.2410)$$

$$(\mathbf{b} \cdot \mathbf{p}, |\mathbf{p}|) \rightarrow (+0.8300, +1.2410)$$

3.5 Block: $\text{dot}[(1, 1)] \times \text{mag}[(1, 0)]$

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 4

Encoded values: +2.3300, +2.5216, -0.3392, +2.9868

Decoded pairs (u, v) :

$$(\mathbf{a} \cdot \mathbf{p}, |\mathbf{b}|) \rightarrow (+1.5000, +1.1874)$$

$$(\mathbf{b} \cdot \mathbf{p}, |\mathbf{a}|) \rightarrow (+0.8300, +1.3342)$$

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 4

Encoded values: +2.3300, +2.5216, -0.3392, +2.8885

Decoded pairs (u, v) :

$$(\mathbf{a} \cdot \mathbf{p}, |\mathbf{a}|) \rightarrow (+1.5000, +1.3342)$$

$$(\mathbf{b} \cdot \mathbf{p}, |\mathbf{b}|) \rightarrow (+0.8300, +1.1874)$$

3.6 Block: $\text{dot}[(2, 0)] \times \text{dot}[(2, 0)]$

PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (2, 0), type=null_self, kind=NULL_SELF, encoded dim= 0

Decoded pairs (u, v) :

$$(\mathbf{a} \cdot \mathbf{b}, \mathbf{a} \cdot \mathbf{b}) \rightarrow (+0.9100, +0.9100)$$

3.7 Block: $\text{dot}[(2, 0)] \times \text{eps}[(2, 1)]$

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= (2, 0), type=12, kind=ASSOC, encoded dim= 2

Encoded values: +1.8200, +1.2851

Decoded pairs (u, v) :

$$(\mathbf{a} \cdot \mathbf{b}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (+0.9100, +0.6760)$$

$$(\mathbf{a} \cdot \mathbf{b}, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (+0.9100, -0.6760)$$

3.8 Block: dot[(2, 0)] $\times$ mag[(0, 1)]

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=2 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 2

Encoded values: +0.9100, +1.2410

Decoded pairs (u, v) :

$$(\mathbf{a} \cdot \mathbf{b}, |\mathbf{p}|) \rightarrow (+0.9100, +1.2410)$$

3.9 Block: dot[(2, 0)] $\times$ mag[(1, 0)]

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 4

Encoded values: +1.8200, +2.5216, -0.7561, +2.2947

Decoded pairs (u, v) :

$$(\mathbf{a} \cdot \mathbf{b}, |\mathbf{a}|) \rightarrow (+0.9100, +1.3342)$$

$$(\mathbf{a} \cdot \mathbf{b}, |\mathbf{b}|) \rightarrow (+0.9100, +1.1874)$$

3.10 Block: eps[(2, 1)] $\times$ eps[(2, 1)]

PairFlavour overlap=(2, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (2, 1), type=null_self, kind=NULL_SELF, encoded dim= 0

Decoded pairs (u, v) :

$$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (+0.6760, +0.6760)$$

$$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (-0.6760, -0.6760)$$

3.11 Block: eps[(2, 1)] $\times$ mag[(0, 1)]

PairFlavour overlap=(0, 1) type=21 kind=ASSOC dim=2 **PairFlavour** overlap= (0, 1), type=21, kind=ASSOC, encoded dim= 2

Encoded values: +2.4819, -1.9970

Decoded pairs (u, v) :

$$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), |\mathbf{p}|) \rightarrow (-0.6760, +1.2410)$$

$$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), |\mathbf{p}|) \rightarrow (+0.6760, +1.2410)$$

3.12 Block: eps[(2, 1)] $\times$ mag[(1, 0)]

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 4

Encoded values: +0.0000, +2.5216, -2.0412, -0.0992

Decoded pairs (u, v) :

$$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), |\mathbf{a}|) \rightarrow (+0.6760, +1.3342)$$

$$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), |\mathbf{b}|) \rightarrow (-0.6760, +1.1874)$$

3.13 Block: mag[(0, 1)] $\times$ mag[(0, 1)]

PairFlavour overlap=(0, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (0, 1), type=null_self, kind=NULL_SELF, encoded dim= 0

Decoded pairs (u, v) :

$(|\mathbf{p}|, |\mathbf{p}|) \rightarrow (+1.2410, +1.2410)$

3.14 Block: $\text{mag}[(0, 1)] \times \text{mag}[(1, 0)]$

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 4

Encoded values: +2.4819, +2.5216, -0.0442, +3.1292

Decoded pairs (u, v) :

$(|\mathbf{p}|, |\mathbf{a}|) \rightarrow (+1.2410, +1.3342)$

$(|\mathbf{p}|, |\mathbf{b}|) \rightarrow (+1.2410, +1.1874)$

3.15 Block: $\text{mag}[(1, 0)] \times \text{mag}[(1, 0)]$

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=2 **PairFlavour** overlap= (0, 0), type=11_sigma, kind=ASSOC, encoded dim= 2

Encoded values: +3.5661, +3.1900

Decoded pairs (u, v) :

$(|\mathbf{a}|, |\mathbf{b}|) \rightarrow (+1.1874, +1.3342)$

$(|\mathbf{b}|, |\mathbf{a}|) \rightarrow (+1.3342, +1.1874)$

PairFlavour overlap=(1, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (1, 0), type=null_self, kind=NULL_SELF, encoded dim= 0

Decoded pairs (u, v) :

$(|\mathbf{a}|, |\mathbf{a}|) \rightarrow (+1.1874, +1.1874)$

$(|\mathbf{b}|, |\mathbf{b}|) \rightarrow (+1.3342, +1.3342)$

4 Alignment Decoder (recovers the G-orbit of repS evaluations)

—**repS**—: $|\text{repS}| = 5$

—**G**—: $|G| = 2$

Decoded vectors: decoded column vectors: 2

Decoded alignment table (rows=repS atoms, cols=group elements): **Decoded alignment table** (rows= repS atoms, columns= $g \in G$):

Table 1: Decoded alignment table

Atom	g_0	g_1
$ \mathbf{a} $	+1.1874	+1.3342
$ \mathbf{p} $	+1.2410	+1.2410
$\mathbf{a} \cdot \mathbf{b}$	+0.9100	+0.9100
$\mathbf{a} \cdot \mathbf{p}$	+0.8300	+1.5000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$	-0.6760	+0.6760

Ground truth (direct evaluation of $g \cdot \text{repS}$ on event): **Ground truth** (direct evaluation of $g \cdot \text{repS}$ on event):

Table 2: Ground truth alignment table

Atom	g_0	g_1
$ \mathbf{a} $	+1.3342	+1.1874
$ \mathbf{p} $	+1.2410	+1.2410
$\mathbf{a} \cdot \mathbf{b}$	+0.9100	+0.9100
$\mathbf{a} \cdot \mathbf{p}$	+1.5000	+0.8300
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$	+0.6760	-0.6760

Multiset comparison (decoded vs ground truth): **Multiset comparison:**

MATCH (multisets equal, atol= 10^{-8})

Max —error— per repS row (decoded vs ground truth): **Max |error| per repS row:**

mag(a): $|\mathbf{a}|$: max err = $0.00e+00$

mag(p): $|\mathbf{p}|$: max err = $0.00e+00$

dot(a,b): $\mathbf{a} \cdot \mathbf{b}$: max err = $0.00e+00$

dot(a,p): $\mathbf{a} \cdot \mathbf{p}$: max err = $0.00e+00$

eps(a,b,p): $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$: max err = $0.00e+00$

5 Summary

Phase 1 encoded length: Phase 1 encoded length: 7 reals

Phase 2 encoded length: Phase 2 encoded length: 42 reals

Total encoded length: Total encoded length: 49 reals

repS size: $|\text{repS}| = 5$ atoms

—G—: $|G| = 2$ group elements

Decoded vectors: decoded alignment vectors: 2

TeX output written to: /Users/lester/github/Echinocoder/symcoder/demo_roundtrip.tex Compile with: pdflatex demo_roundtrip.tex